Chapter 10 formulas

1. Simple Interest
   - $F = P(1 + rt)$
   - $P = \dfrac{F}{1 + rt}$
   - $r = \dfrac{\frac{F}{P} - 1}{t}$
   - $t = \dfrac{\frac{F}{P} - 1}{r}$
2. Compound Interest
   - $F = P(1 + p)^T$
   - $P = \dfrac{F}{(1 + p)^T}$
   - $p = \left(\dfrac{F}{P}\right)^{1/T} - 1$
   - $T = \dfrac{\log(\frac{F}{P})}{\log(1 + p)}$ (Note: this is not the same as $T = \dfrac{\frac{F}{P}}{(1 + p)}$)
3. Amortization Formula
   - $M = P\dfrac{p(1 + p)^T}{(1 + p)^T - 1}$
   - $P = M\dfrac{(1 + p)^T - 1}{p(1 + p)^T}$
   - $T = \dfrac{-\log\left(1 - \frac{pP}{M}\right)}{\log(1 + p)}$
4. Annuity Formula
   - $F = M\dfrac{(1 + p)^T - 1}{p}$
   - $M = F\dfrac{p}{(1 + p)^T - 1}$
   - $T = \dfrac{\log\left(1 + \frac{pF}{M}\right)}{\log(1 + p)}$